Monte Carlo Methods for Estimation: Recap

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Module 1: Integrate — Session 3

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One draw is now a whole dataset

An estimator is a function of the sample, so it is a **random variable**. Its distribution is the **sampling distribution**, and every question we ask about an estimator is an expectation over it.

```
theta_hats <- replicate(k, {
  x <- rmodel(n, theta)    # simulate a whole dataset of size n
  estimator(x)             # collapse it to one number
})
```

- n is the usually given by the a real data set
- k is your our choice: time vs. precision of estimate

Bias, variance, and MSE

$$\text{Bias}(\hat{\theta}) = \text{E}\left(\hat{\theta}\right) - \theta, \qquad \text{MSE}(\hat{\theta}) = \text{E}\left((\hat{\theta} - \theta)^2\right) = \text{Var}\left(\hat{\theta}\right) + \text{Bias}(\hat{\theta})^2$$

- All three are expectation, estimate using mean of $\hat{\theta} - \theta$, or $(\hat{\theta} - \theta)^2$.
- MSE balance accuracy and precision
- The ranking can depend on n, so **try several** n

Coverage and Expected Width

$$\text{coverage} = P(A \leq \theta \leq B) = \mathrm{E}\left(\mathbf{1}(A \leq \theta \leq B)\right)$$

- Replicate samples with known θ, create CI for each sample, proportion that include true θ
- Coverage is a proportion out of k: report `binom.test(hits, k)$conf.int` alongside it.
- After establishing coverage, compare different CI methods on **expected width** $\mathrm{E}\,(B - A)$.